

LU Junyu

Ph.D. in Mathematics

Low-dimensional topology and left-orderable groups

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📍 Xiamen, Fujian, China

Employment History

- 2019.01–2019.04 📌 **Trading Systems Software Engineer**, Hainan Xinruan Software Co., Ltd., Beijing Branch, Beijing, China.
Maintained and enhanced a transaction matching system, including code updates, testing, and troubleshooting. Collaborated with the development team to improve system reliability and maintainability.
- 2018.07–2018.10 📌 **Algorithm Engineer**, Beijing Xiaolian (MultiVAC) Technology Co., Ltd., Beijing, China.
Implemented research algorithms as production-oriented software components for a blockchain platform. Developed reusable interfaces for system integration and supported algorithm testing and optimization.
- 2014.08–2015.08 📌 **Project Officer**, Nanyang Technological University, Singapore.
Conducted tutorial classes for first- and second-year undergraduate courses. Prepared and refined tutorial materials, explained mathematical concepts and problem-solving techniques, and provided individual academic support during student consultations. Liaised between students and course instructors by communicating common learning difficulties and student feedback.
- 2013.01–2014.01 📌 **Project Officer**, Nanyang Technological University, Singapore.

Education

- 2021.01–2026.06 📌 **Ph.D. in Mathematics**, University of Manitoba, Winnipeg, Canada.
Supervisor: Adam Clay.
Thesis: *Order Detection and Left-orderability of 3-Manifold Groups*.
- 2016.09–2018.06 📌 **M.A. in Mathematics**, McGill University, Montreal, Canada.
Supervisors: Marcin Sabok and Piotr Przytycki.
Thesis: *On Convex Normal Subgroups*.
- 2008.08–2013.02 📌 **B.Sc. in Physics, First Class Honours**, Nanyang Technological University, Singapore.

Scholarships and Fellowships

- 2024–2025  **Dr. Jiri Sichler Memorial Scholarship in Algebra**, University of Manitoba; awarded for strong academic standing and graduate study with a focus on algebra.
- 2021–2026  **Fellowship for Education Purposes**, University of Manitoba; graduate fellowship funding supporting full-time doctoral study and research in Mathematics.

Research Interests

-  Low-dimensional topology and geometry
-  Left-orderable groups and orderability of 3-manifold groups
-  L-space conjecture

Publications and Preprints

- 1 A. Clay and **J. Lu**, “Order-detection, representation-detection, and applications to cable knots”, *Transactions of the American Mathematical Society* **379**(3) (2026), 1755–1798. doi:10.1090/tran/9525.
- 2 A. Clay and **J. Lu**, “Order-detection and non-left-orderable surgeries on links”, revised manuscript submitted to *Pacific Journal of Mathematics*, 20 pp. arXiv:2505.01564.
- 3 A. Clay, **J. Lu**, and T. Staruch, “A combinatorial approach to obstructing left-orderability of Dehn fillings”, preprint, 21 pp. arXiv:2607.15603.

Research and Seminar Talks

- “3-Manifold Groups ABC”, Geometry and Topology Learning Seminar, University of Manitoba, 2025.
- “Order-detected Slopes on Cable Knots”, Graduate Research Talk, Redbud Conference, Spring 2024.
- “A Step Towards Character Varieties and A-polynomials”, Geometry and Topology Learning Seminar, University of Manitoba, 2024.
- “A Layman’s Introduction to Knots and Jones Polynomials”, Geometry and Topology Learning Seminar, University of Manitoba, 2023.
- “Left Orderability and 3-Manifold Groups: A Short Invitation”, Graduate Mathematics Seminar, University of Manitoba, 2023.
- “On Convex Normal Subgroups”, Geometric Group Theory Seminar, McGill University, 2018.

Teaching Experience

Instructor, University of Manitoba

- MATH 2170, *Number Theory 1*, Winter 2024. Independent course preparation; approximately 20 students.
- MATH 1500, *Introduction to Calculus*, Winter 2023. Coordinated course taught by four instructors; approximately 150 students.

Teaching Assistant

- University of Manitoba: tutorials and student support for MATH 1500, MATH 1524, MATH 1240, MATH 2040, MATH 2080, MATH 2090, and other first- and second-year mathematics courses.
- McGill University: MATH 141, *Calculus 2*, Winter 2017.
- Nanyang Technological University: tutorials for various first- and second-year calculus and algebra courses, 2013–2015.